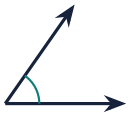
Lesson 1.5

Angles

Measure, classify, add — and the angle pairs

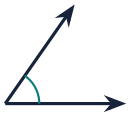
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Four kinds of angles

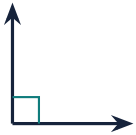


Acute
less than 90°

Four kinds of angles

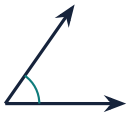


Acute
less than 90°

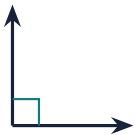


Right
exactly 90°

Four kinds of angles



Acute
less than 90°

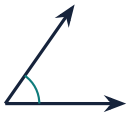


Right
exactly 90°

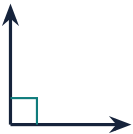


Obtuse
between 90° and 180°

Four kinds of angles



Acute
less than 90°



Right
exactly 90°

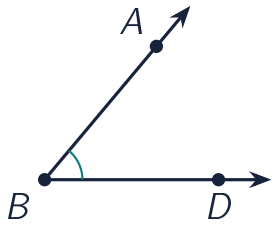


Obtuse
between 90° and 180°



Straight
exactly 180°

Reading an angle's name

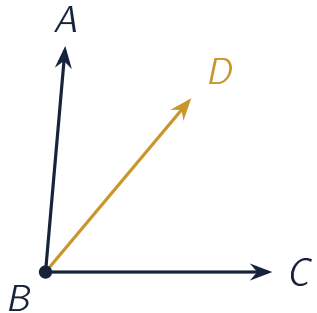


$\angle ABD$

The **vertex** is the **middle** letter: B.

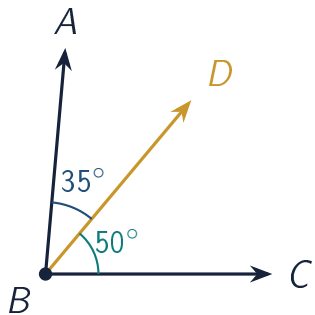
$\angle ABD$ and $\angle DBA$ name the same angle.

Angle Addition Postulate



D is inside $\angle ABC$.

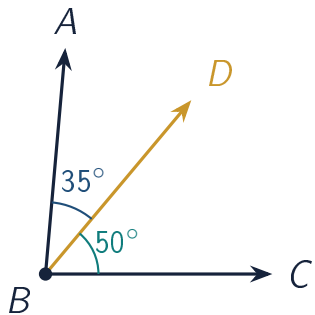
Angle Addition Postulate



D is inside $\angle ABC$.

$m\angle ABD = 35°,\ m\angle DBC = 50°$

Angle Addition Postulate



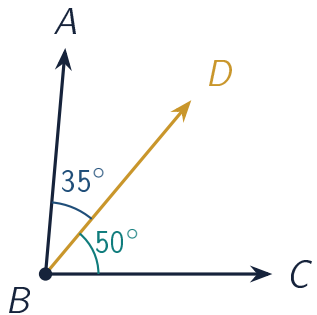
D is inside $\angle ABC$.

$m\angle ABD = 35°,\ m\angle DBC = 50°$

$m\angle ABC = 35 + 50 = 85°$

the two small angles fill the big one

Angle Addition Postulate



D is inside $\angle ABC$.

$m\angle ABD = 35°,\ m\angle DBC = 50°$

$m\angle ABC = 35 + 50 = 85°$

the two small angles fill the big one

$85° < 90°$: **acute**

With algebra

D is inside $\angle ABC$.
$m\angle ABD = 2x + 5, \quad m\angle DBC = 3x, \quad m\angle ABC = 80^\circ$.

Find $m\angle DBC$.

With algebra

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$m\angle ABD = 2x + 5, \quad m\angle DBC = 3x, \quad m\angle ABC = 80^\circ.$

Find $m\angle DBC$.

$(2x + 5) + 3x = 80$ part + part = whole

With algebra

D is inside $\angle ABC$.
$m\angle ABD = 2x + 5, \quad m\angle DBC = 3x, \quad m\angle ABC = 80^\circ.$

Find $m\angle DBC$.

$(2x + 5) + 3x = 80$ part + part = whole
$5x + 5 = 80 \Rightarrow 5x = 75 \Rightarrow x = 15$

With algebra

D is inside $\angle ABC$.
$m\angle ABD = 2x + 5, \quad m\angle DBC = 3x, \quad m\angle ABC = 80^\circ.$

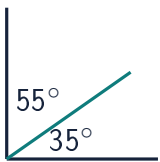
Find $m\angle DBC$.

$(2x + 5) + 3x = 80$ part + part = whole

$5x + 5 = 80 \Rightarrow 5x = 75 \Rightarrow x = 15$

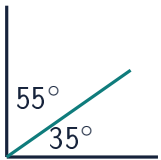
$m\angle DBC = 3(15) = 45^\circ$ check: $35 + 45 = 80$ ✓

Angle pairs

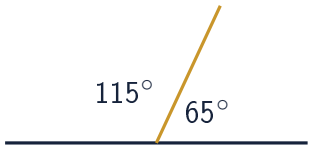


Complementary: add to 90°

Angle pairs

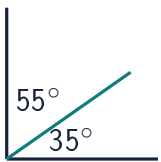


Complementary: add to 90°

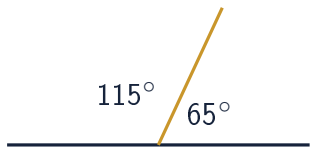


Supplementary: add to 180°

Angle pairs



Complementary: add to 90°



Supplementary: add to 180°

C comes before **S** in the alphabet — just like 90 comes before 180.

Find the missing angle

Supplement of $65°$?

Find the missing angle

Supplement of $65°$?

$180 - 65 = 115°$

Find the missing angle

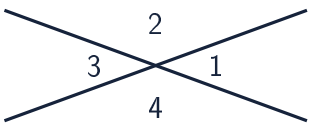
Supplement of $65°$?

$180 - 65 = 115°$

Complement of $65°$?

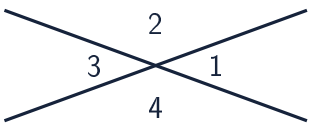
$90 - 65 = 25°$

Vertical angles and linear pairs



Vertical angles are across from each other, and they are **equal**: $\angle 1 = \angle 3$.

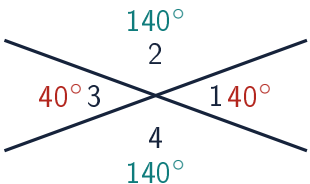
Vertical angles and linear pairs



Vertical angles are across from each other, and they are **equal**: $\angle 1 = \angle 3$.

Linear pair: side by side on a line, they add to $180°$: $\angle 1 + \angle 2 = 180°$.

Vertical angles and linear pairs



Vertical angles are across from each other, and they are **equal**: $\angle 1 = \angle 3$.

Linear pair: side by side on a line, they add to 180°: $\angle 1 + \angle 2 = 180°$.

If $\angle 1 = 40°$: $\angle 3 = 40°$, $\angle 2 = \angle 4 = 140°$.

Your turn! Pause the video and try it.

$\angle 1$ and $\angle 2$ are supplementary.
$m\angle 1 = 4x + 20, \quad m\angle 2 = 2x + 40.$

Find x and both angles.

Your turn! Pause the video and try it.

$\angle 1$ and $\angle 2$ are supplementary.
$m\angle 1 = 4x + 20, \quad m\angle 2 = 2x + 40.$
Find x and both angles.

$$(4x + 20) + (2x + 40) = 180 \Rightarrow 6x + 60 = 180 \Rightarrow x = 20$$

$m\angle 1 = 100^\circ, \quad m\angle 2 = 80^\circ \qquad 100 + 80 = 180 \checkmark$

Remember

- Acute $< 90°$, right $= 90°$, obtuse between, straight $= 180°$.
- The vertex is the **middle** letter. Small angles add up to the big one.
- **Complementary** $90°$, **supplementary** $180°$, **vertical** angles are equal.

Now practice at your level:

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